Types of AI Models

Rule-based, Learning-based, Supervised, Unsupervised and Deep Learning

Your Name

Your School / Institute

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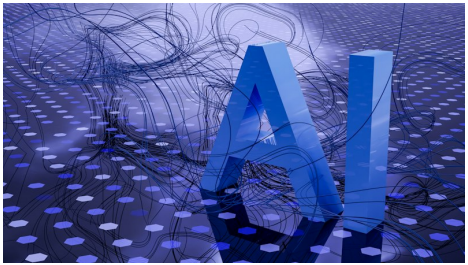
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Learning Outcomes

Students will be able to:

- distinguish rule-based and learning-based AI models
- explain supervised, unsupervised and reinforcement learning
- understand regression and classification
- describe deep learning and neural networks
- choose a suitable model type for a given problem



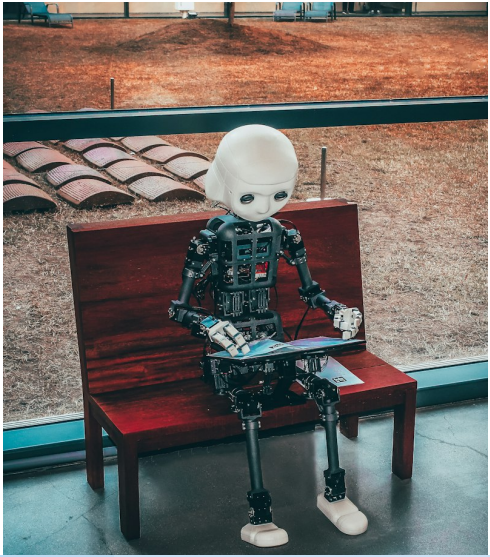
Introduction to AI Models

What Is an AI Model?

- A mathematical representation that learns patterns from data
- Maps inputs to outputs (predictions or decisions)
- Machines need data as numbers (0s and 1s)
- Humans understand graphs; models use equations

Core idea

AI models define relationships between parameters mathematically.



Two Broad Approaches

Rule-based

Developer writes explicit rules.
"If this, then that."

Learning-based

Machine learns rules from data.
No hand-coded logic needed.

Most modern AI systems are learning-based.

Rule-based Models

Rule-based Approach

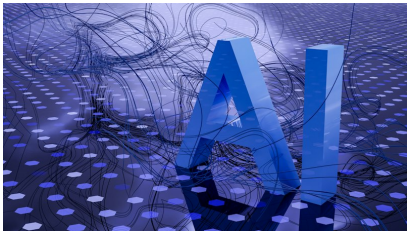
- Developer feeds data + rules
- Example: identify banana vs orange from 200 labelled images
- Labels act as rules
- Cannot recognise unseen categories (e.g. an apple)

Typical use

Simple expert systems, fixed decision trees, early AI.

Limitations of Rule-based Models

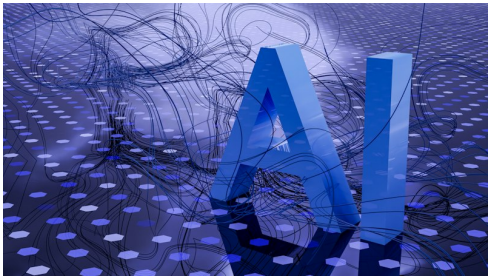
- Hard to write rules for complex problems
- Does not improve with more data
- Brittle – fails on new situations
- Maintenance becomes expensive



Learning-based Models

Learning-based Approach

- Machine learns on its own from data
- Works with labelled and unlabelled data
- Discovers patterns and defines its own rules
- Example: cluster 1000 tree images by height, colour, fruit, leaves



Rule-based vs Learning-based

Rule-based	**Learning-based**
Follows developer rules	Learns from data
Needs labelled data mostly	Handles labelled + unlabelled
Less training time	More training time
Hard to scale	Scales with data

Supervised Learning

Supervised Learning

- Trained on a labelled dataset
- Inputs come with known correct outputs
- Data split into training and test sets
- Fast training because answers are known
- After training, predicts on new inputs



Training vs Testing Data

Training data

Used to teach the model. Must be relevant, accurate and sufficient.

Testing data

Unseen data used to evaluate performance. Represents real-world cases.

Regression

- Predicts continuous numerical values
- Examples: house price, salary, temperature
- Learns a function (often a line) from data
- Predicts future values from past patterns

Example

Predict tomorrow's temperature from humidity and cloud cover.



Classification

- Predicts discrete categories or labels
- Examples: spam / not spam, cat / dog, disease / no disease
- Output is a class, not a number

Common algorithms

Logistic regression, Decision trees, Random forest, SVM, Neural networks



Supervised Learning Examples

- Email spam detection
- Credit risk scoring
- Image recognition (labelled photos)
- Medical diagnosis from test results
- Predicting exam scores

Unsupervised Learning

Unsupervised Learning

- Training data is neither classified nor labelled
- Model finds hidden structure by itself
- Discovers groups, patterns or anomalies
- Once learned, can apply to new data



Clustering

- Groups similar items together
- Example: customers with similar buying behaviour
- Common method: K-means
- Number of groups may be chosen by the user

Textbook example

Images of cycle, car and helicopter can be separated into groups, but the model cannot name the groups because data is unlabelled.

Dimensionality Reduction & Association

Dimensionality reduction

Reduce number of features while keeping important information.
Example: PCA

Association rules

Find relationships between items.
Example: “people who buy bread also buy butter”

Unsupervised Learning Examples

- Customer segmentation
- Anomaly detection (fraud, faults)
- Topic modelling in documents
- Organising large photo collections
- Market basket analysis

Other Learning Types

Semi-supervised Learning

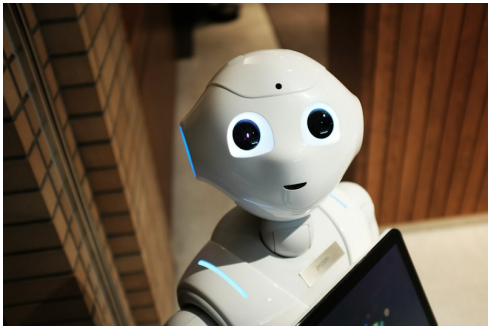
- Uses a small amount of labelled data + large amount of unlabelled data
- Useful when labelling is expensive
- Combines strengths of supervised and unsupervised approaches

Practical benefit

High accuracy with less labelling effort.

Reinforcement Learning

- Agent learns by trial and error
- Receives rewards or penalties
- Goal: maximise long-term reward
- Examples: games, robotics, self-driving cars



Reinforcement Learning Loop

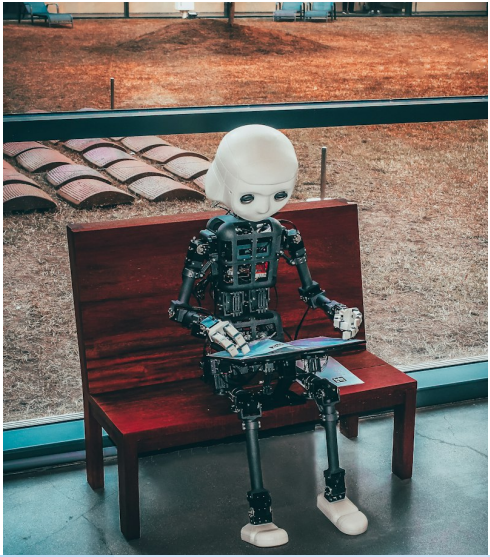


The agent improves its policy over many interactions.

Deep Learning

Deep Learning

- Subset of machine learning
- Uses multi-layer neural networks
- Mimics the human brain
- Learns hierarchical features from large data
- Example: driverless car identifying a person crossing the road



Neural Networks Basics

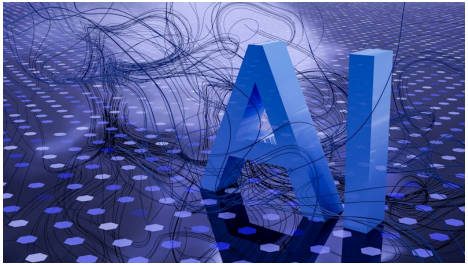
- Made of interconnected nodes (neurons)
- Layers: input $\rightarrow$ hidden $\rightarrow$ output
- Each connection has a weight
- Learns by adjusting weights

Deep network

Many hidden layers $\rightarrow$ can learn complex patterns.

Common Deep Learning Architectures

- Convolutional Neural Networks (CNNs) – images
- Recurrent Neural Networks (RNNs) – sequences
- Transformers – language and vision
- Generative models (GANs, Diffusion) – create new data



When to Use Deep Learning

- Large amounts of data available
- Complex patterns (images, speech, text)
- High accuracy is required
- Computational resources are sufficient

Note

Not always the best choice for small tabular datasets.

Choosing a Model

How to Choose a Model Type

Labelled data available? → Supervised

Need continuous number? → Regression

Need category? → Classification

No labels? → Unsupervised / Clustering

Learn from actions? → Reinforcement

Complex data (images/text)? → Deep Learning

Comparison Summary

Type	Needs labels?	Typical output
Supervised	Yes	Prediction / class
Unsupervised	No	Groups / structure
Reinforcement	Rewards	Actions / policy
Deep Learning	Usually yes	Complex patterns

Real-world Mapping

- Spam filter → Supervised classification
- House price prediction → Supervised regression
- Customer groups → Unsupervised clustering
- Game AI / robot → Reinforcement learning
- Face recognition → Deep learning (CNN)
- Chatbot → Deep learning (Transformer)

Summary

Key Takeaways

- Rule-based: explicit rules written by humans
- Learning-based: model learns from data
- Supervised: labelled data, predicts known targets
- Unsupervised: finds hidden structure without labels
- Reinforcement: learns by reward and trial
- Deep learning: multi-layer neural networks for complex data

Quick Recap Table

Model type	**One-line description**
Rule-based	Follows hand-written rules
Supervised	Learns from labelled examples
Unsupervised	Discovers patterns without labels
Reinforcement	Learns optimal actions via rewards
Deep Learning	Neural networks with many layers

Practice Question

Which model type would you use?

1. Predict rainfall amount tomorrow
2. Group students by learning style (no labels given)
3. Train a robot to walk
4. Detect cats in photos

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Which model type would you use?

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Answers: 1. Regression 2. Clustering 3. Reinforcement 4. Deep learning / CNN

Further Reading

- Supervised vs Unsupervised learning
- Evaluation metrics (accuracy, precision, recall)
- Overfitting and underfitting
- Popular libraries: scikit-learn, TensorFlow, PyTorch

Thank You!

Questions on Types of AI Models?

